

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**

ORGANISATION OF ISLAMIC COOPERATION (OIC)

**Department of Computer Science and Engineering (CSE)**

SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2024-2025

DURATION: 2 HOURS

FULL MARKS: 120

**SWE 4101: Introduction to Software Engineering****Programmable calculators are not allowed. Do not write anything on the question paper.**Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) A software company is developing a new information management system for a large organization. To determine which features and services the system should provide, the development team conducts several requirement-gathering activities. First, the company invites several experienced professionals who have deep knowledge of the business domain. These experts participate in group discussions, freely sharing ideas, suggesting possible solutions, and building on one another's opinions to solve complex problems related to the proposed system. Next, the analysts arrange structured discussions with a small group of client representatives and potential users. During these sessions, participants share their expectations, opinions, and concerns about the new system to help the development team clearly understand the needs of the target users. The analysts also spend time watching how employees currently perform their daily tasks using the existing system in their workplace. This helps the team understand the actual workflow and identify practical requirements. In addition, the development team prepares diagrams that show how different users will interact with the system to perform specific tasks. They also analyze how the new software will communicate and exchange information with other existing systems and user interfaces. Based on the scenario above, mention and explain the requirement gathering tools and techniques used by the company.
- Brainstorming  
Focus group  
Use case diagram*
- Observation*
- Interface Analysis*
- b) Maintainability, dependability, efficiency, and acceptability are the four core attributes of good software. Explain the maintainability and dependability of a software product with suitable examples.
- c) Absolute Broke Bank Ltd. (ABBL), wants to develop a Mobile wallet for its clients. They already have more than a million users, and their service should be up 99.98% time. If the service is unavailable, users should receive a notification immediately. Also, the app should respond within a second to any user request. In any situation, if the app fails to respond, it should provide a proper error message to the user. The app's CPU usage should not exceed 15%, and its memory consumption should be less than 20 MB. They want your company to develop a mobile wallet for them that includes all the existing features of their internet banking, and they also want to add many new features. The new features have not yet been decided. And ABBL wants to observe progress in development every 2 weeks.
- i. Under these circumstances, which Software development model will you choose to complete the development? Briefly mention all the events related to your chosen development model.
- (ii) Identify five non-functional requirements mentioned in the above scenario.
2. a) Briefly explain the five fundamental feasibility evaluations that are crucial for determining a software project's viability, practicality, and strategic alignment before its initiation.
- FLATS* b) Explain the difference between a requirement and its specification with the help of an example.

10  
(CO3)  
(PO2)8  
(CO3)  
(PO1)12 + 5  
(CO3)  
(PO2)10  
(CO3)  
(PO1)  
7  
(CO3)  
(PO1)

Us

- c) The Institute of Unconventional Technology (IUT) provides an online Course Registration System that helps students manage their academic enrollment. To register for courses, a student must first log in to the system using their student ID and password. After logging in, the student can view the list of available courses offered in the current semester. A student may select and enroll in courses based on their program requirements. The system allows students to add new courses, drop previously selected courses, and view their registered course list. The system also checks whether the selected courses conflict in schedule or exceed the maximum allowed credit limit. Students can log in to the system at any time during the registration period to update their course selections or review their registered courses. 8 + 10  
(CO3)  
(PO2)
- Ex: →
- i. Draw a Use Case Diagram for the Course Registration System.
  - ii. Write a Use Case Description for the use case *Register Course*.
3. a) With proper justification, mention the type of maintenance required for the below scenarios: 5 + 5  
(CO3)  
(PO2)
- i. The Institute of Unconventional Technology (IUT) has a mobile application for students where they can check class schedules, exam results, and campus announcements. Recently, the number of students using the app has increased significantly. During busy times, many students experience slow loading and delays when checking their results or schedules. The development team decides to improve the systems performance so that it can handle many users at the same time without slowing down.
  - ii. The Institute of Unconventional Technology (IUT) uses a library management system to keep records of books and student borrowings. The system has been working properly for several years. However, the developers notice that some parts of the system use old security methods and outdated software libraries. Although the system still works, the team decides to update these components to prevent future problems and ensure the system remains secure and reliable.
- b) During a software upgrade, the testing team is responsible for performing both sanity and regression testing. Discuss the difference between sanity and regression testing, and explain how these testing types contribute to ensuring the stability of the software after new changes 10  
(CO3)  
(PO2)
- c) A software application is released with the version number 2.4.7.133. Explain the possible meaning of each segment of this version number in software versioning. 5  
(CO3)  
(PO1)
4. a) "Imagine that the developer who comes after you is a violent psychopath who knows where you live." This quote motivates developers to write clean code. Explain the importance of writing clean code and how it benefits future developers. 5
- b) Briefly explain the aspects that a tester aims to verify during security testing of a software system. 8
- c) How does continuous delivery differ from continuous deployment? 5
- d) What is reverse engineering? Explain the purpose and benefit of refactoring. 2 + 5